

BRAIN and Alive Engine



Around twenty researchers making AI more accessible: less energy, less data, fewer priors on the data. A broad international network, and years of work on large language models and now on agentic AI.



A platform to build, deploy, and supervise persistent AI teammates that learn continuously, accumulate expertise, and work alongside human teams. On-prem for sensitive data, sandboxed, and fully auditable.

Team and roles



Jean-Charles Vialatte

Founder, [Alive Engine](#)

- PhD 2018, graph deep learning, IMT Atlantique
- 2018 to 2025, ML / AI engineer
- 2025, founder and builder of Alive Engine



Bastien Padeloup

Associate Professor, [IMT Atlantique](#)

- PhD 2017, graph signal processing, IMT Atlantique
- Lab-STICC, BRAIN team
- Graph ML, Few-shot AI, LLMs, Healthcare

Talk structure

- 1 Deploying agentic AI
- 2 Emergency stroke care
- 3 Supervising trust



ACT ONE

Deploying agentic AI

How an autonomous AI runs inside the hospital, on its own data, and what that solves.



ON-PREM MODEL

LOCAL GATEWAY

INBOUND ONLY

SECURE MESSAGING

SUB-AGENTS

What is an AI agent?

Equip a model with the tools of the service: databases, files, a shell, search. It stops chatting and starts doing the work.

OPEN MODEL

a capable chat, on its own



THE HARNESS



Databases



Files



Shell



Search

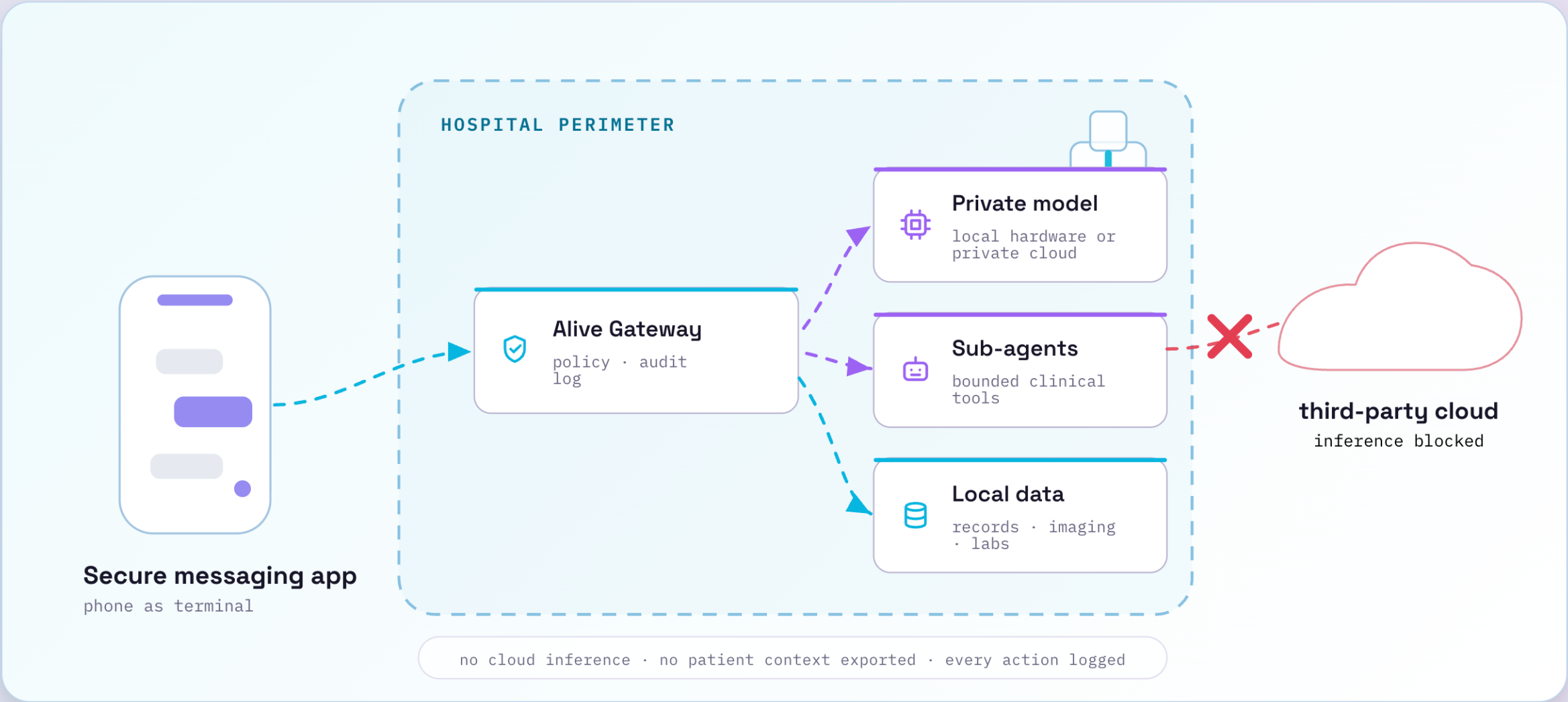


AGENT

does the work

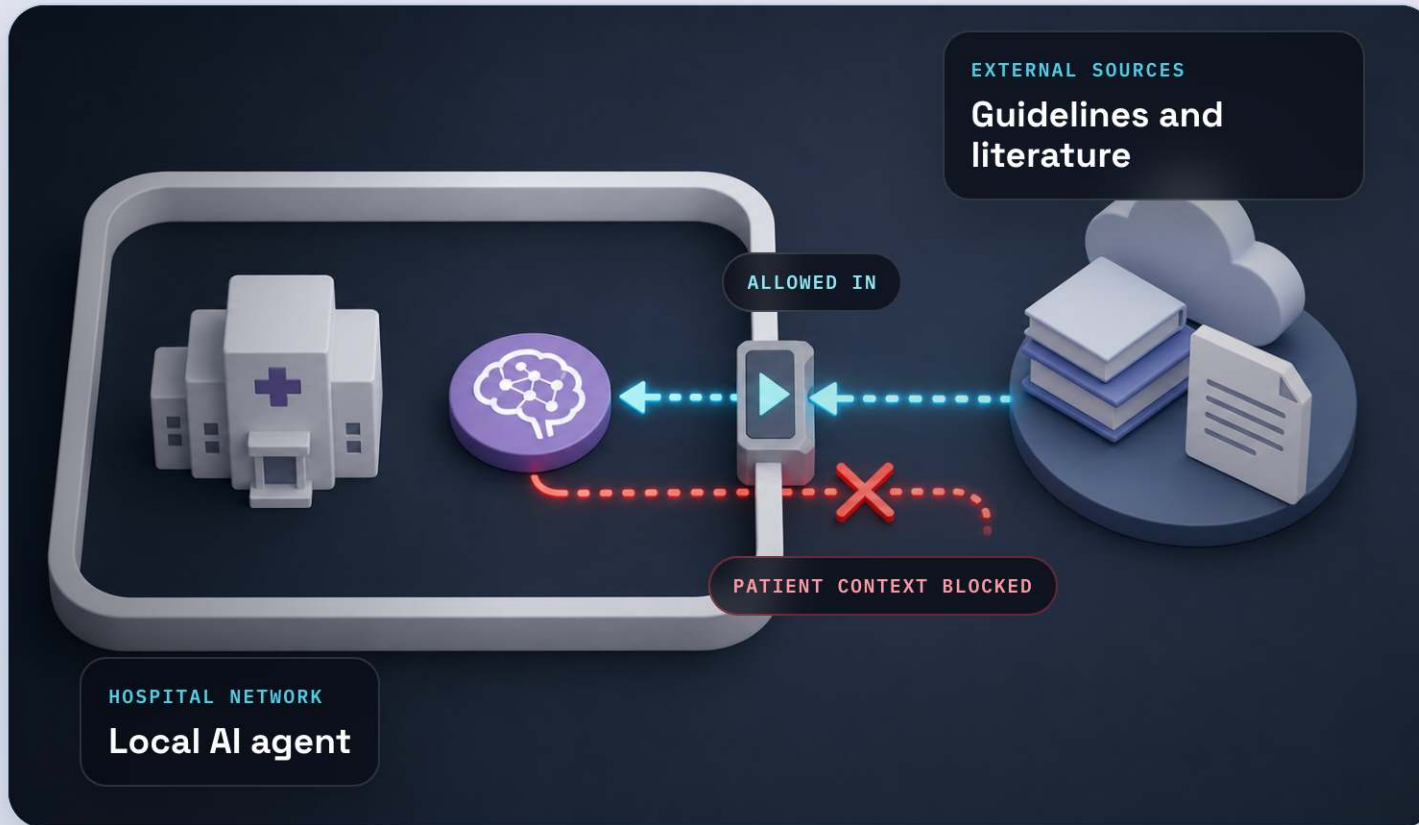
Hospital AI stays inside the wall

The clinician UI, gateway, model, agents, and hospital data all run on the hospital network. Cloud inference is not in the loop.



Patient data never leaves the perimeter

The agent may pull outside knowledge in. It cannot send patient context out to a third-party model.



Allowed

guidelines and literature come into the hospital network

Blocked

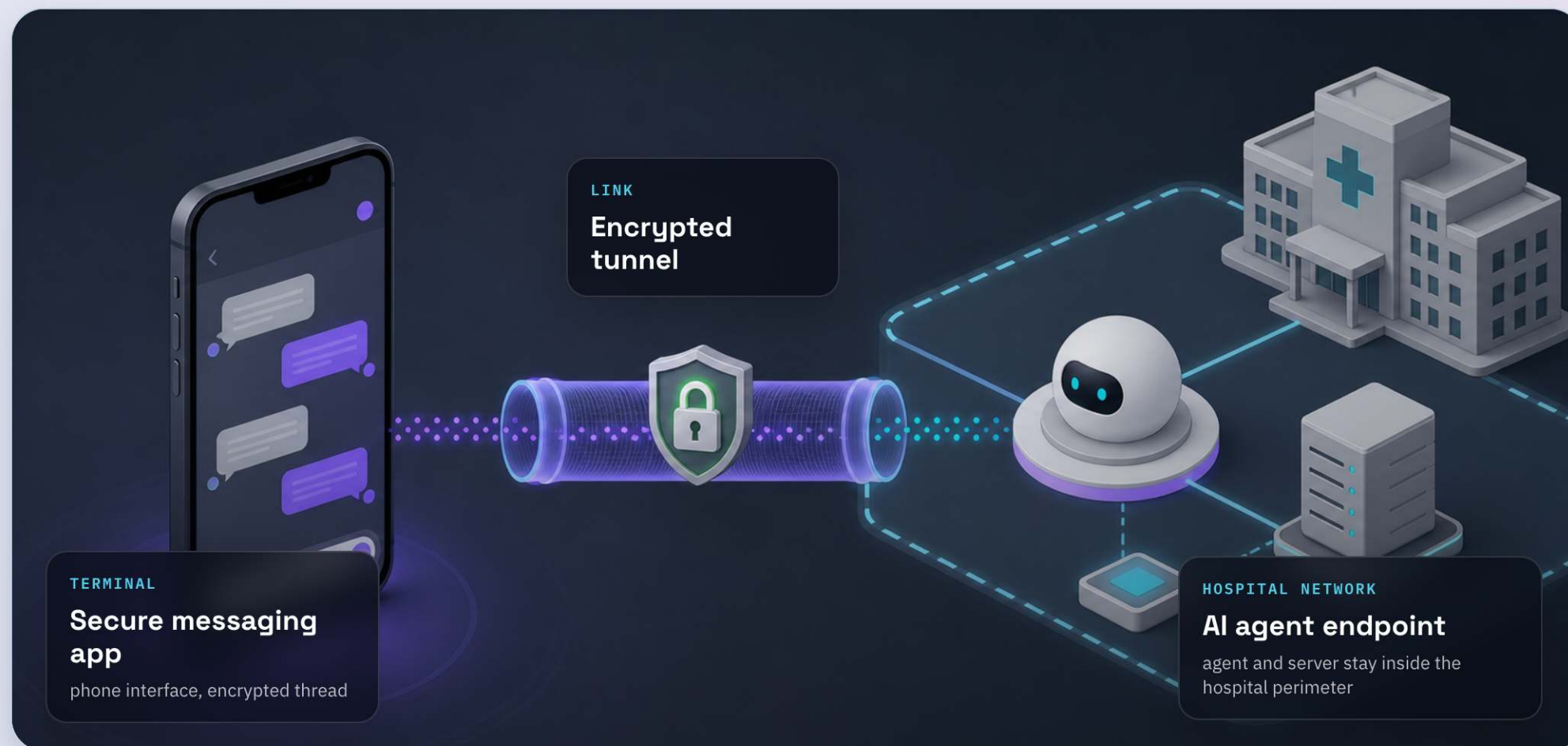
patient context cannot be sent to an external model

Recorded

gateway, model, and tool calls are audit events

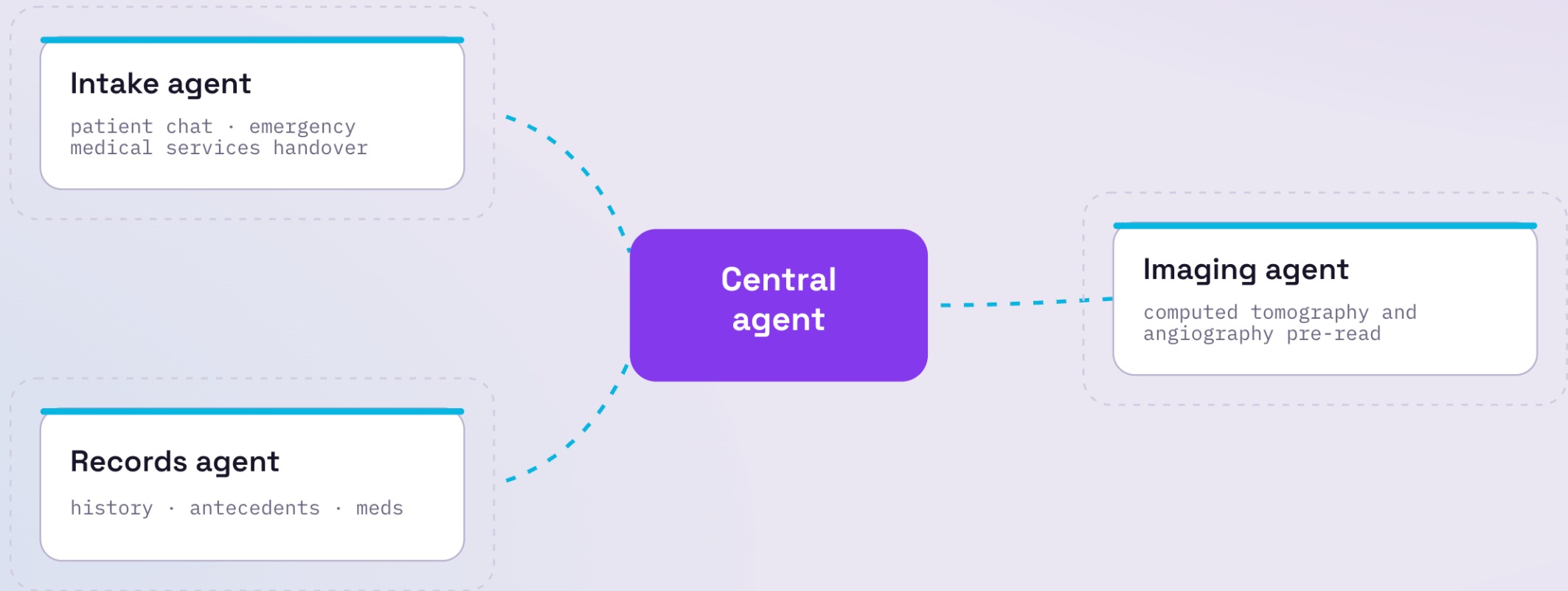
Phone as secure terminal

Clinicians can reach the agent from a secure messaging app. The far end of the thread terminates inside the hospital network.



Sub-agents split the clinical work

A central agent orchestrates focused sub-agents, each with its own context and tools.



01 Parallelism

work happens at once, not in sequence

02 Compartmentalization

bounded contexts do not bleed

03 Specialization

each sub-agent tuned to one task of the service

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ACT TWO

Emergency stroke care

How agentic AI can shorten arrival-to-thrombolysis time in acute stroke.



ARRIVAL TO THROMBOLYSIS

INTAKE

HISTORY

IMAGING PRE-READ

COMPARTMENTS

Every minute matters in acute stroke

The deck's stroke use case is about one measurable clock: time from hospital arrival to thrombolysis.

UNTREATED ISCHEMIC STROKE

1,900,000

neurons lost **per minute**

10 min faster $\approx$ **19,000,000**

fewer neurons exposed to untreated ischemia

PATHWAY CLOCK

Clock hospital arrival to thrombolysis

60 min 90th-percentile target (guideline)

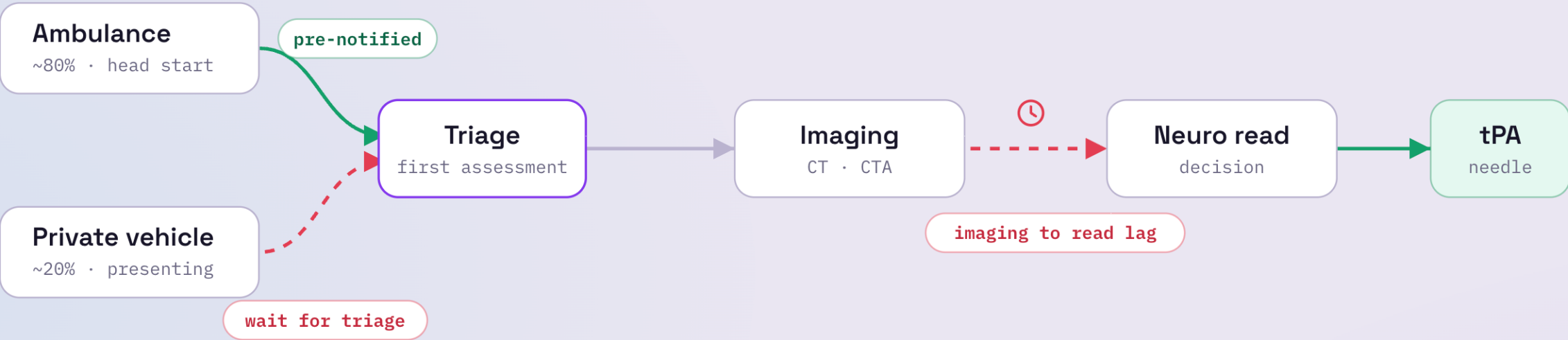
tPA benefit is steeply time-dependent

**The architecture
matters only if it moves
this clock.**

Figures from the stroke-pathway literature (Bulmer et al., Front. Neurol. 2021).

Where the minutes leak

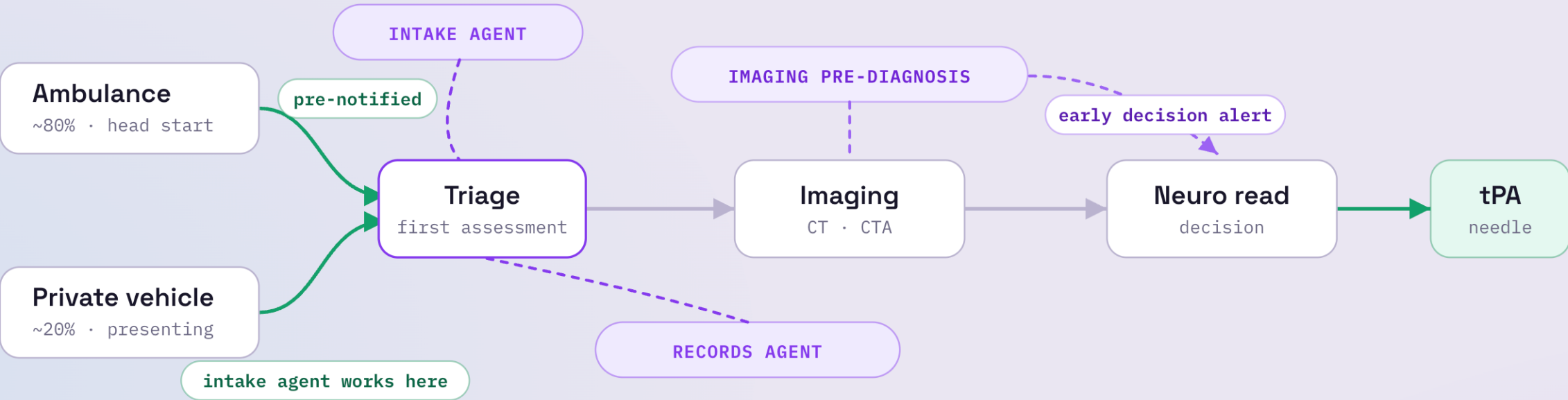
Patients who arrive by ambulance get a head start. Self-presenting patients wait for triage. And a system delay sits between imaging and the neuro read.



Intake can start before triage

On a suspected stroke, the central agent spawns an intake sub-agent. By ambulance? It structures the handover. By private vehicle? It opens a secure thread with the patient or a companion, before triage.

Intake agent · collects onset, symptoms, contraindications before the patient is triaged
Records agent · pulls antecedents from database and cross-checks with intake



WHY AGENTIC Parallel intake collapses the pre-triage and registration gap the literature flags as slow for private-vehicle arrivals.

Records retrieval runs in parallel

A records agent queries the hospital databases and returns a structured brief while the patient is still arriving.

RECORDS AGENT • BRIEF

prior TIA (2023)

apixaban, anticoagulant

FLAG

eGFR normal, no allergies

CONTRAINDICATION

Anticoagulation surfaced before the clinician sits down.

WHY AGENTIC

History retrieval runs **concurrently** with intake and imaging prep, not serially after the clinician arrives.

Pre-read starts when images land

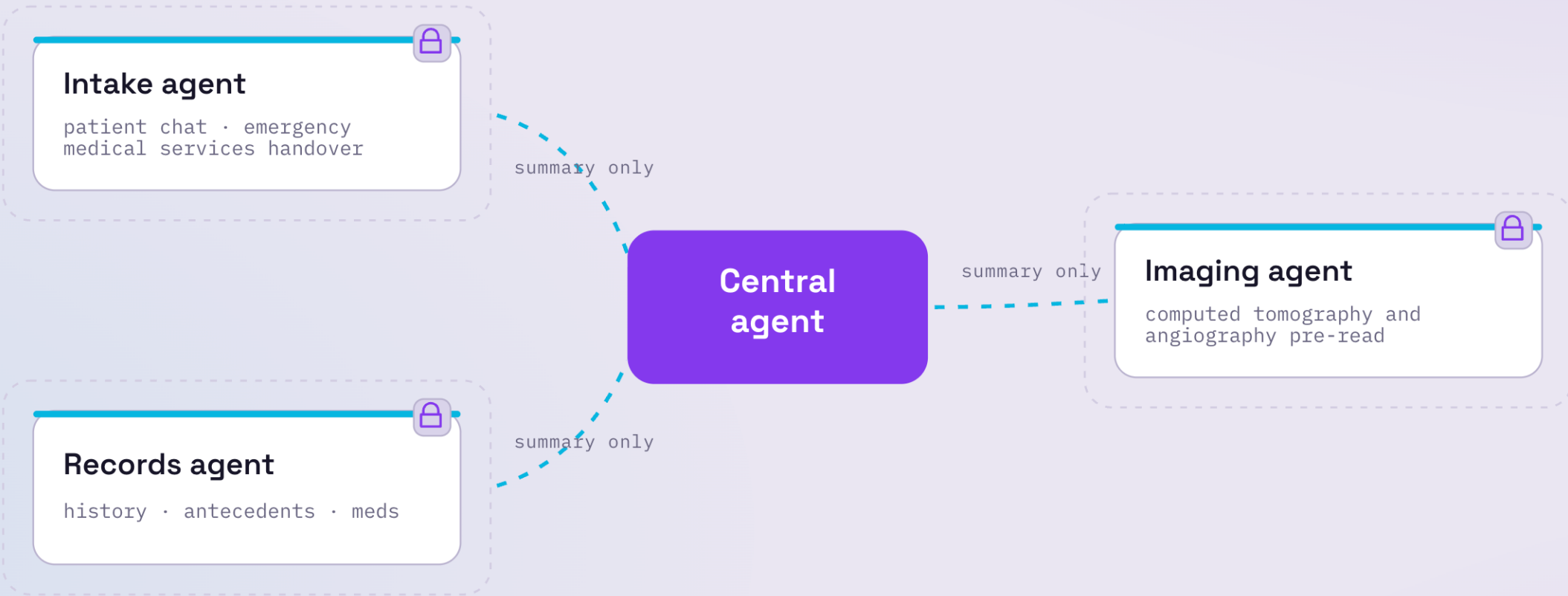
An imaging agent drafts a preliminary read immediately, raises urgency, then the neurologist confirms or overrides.



WHY AGENTIC It does not replace the neurologist. It removes the dead time before the neurologist looks, then a clinician confirms.

Architecture is the privacy control

Each contact point is a separate sub-agent with a bounded context.
Information passes up as structured summaries, never raw context.



WHY AGENTIC

The patient-facing agent never sees the database. The records agent never sees the chat. Smaller contexts, smaller blast radius, full audit.

Agents reduce arrival-to-thrombolysis delay

Agents attack the pre-triage gap, retrieve history in parallel, and cut the imaging-to-read dead time. Minutes saved from hospital arrival to thrombolysis become neurons saved.



≈24.7M

neurons preserved per case, at the projected saving

-13 min × 1.9M neurons/min

Anchor: combining process changes alone cut median arrival-to-thrombolysis time by up to 26.7% in simulation (Bulmer et al., 2021). Agentic gains shown are a projected concept, not a measured result.

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ACT THREE

Supervising trust in medical agents

A funded research project on reliability, traceability, drift monitoring, and validation for medical agents.



PROBLEM

PROJECT STRUCTURE

STATE OF THE ART

SPECIFIC OBJECTIVES

Agentic AI is arriving before trust is solved

The research project starts from a simple risk: in medicine, autonomous systems must be supervised for what they do, not only for what they say.



61%

healthcare organizations already
building or deploying agentic AI



91.8%

clinicians already exposed to medical
hallucinations from foundation models



64–72%

residual hallucinations caused by
causal or temporal reasoning failures

The research question is continuous trust supervision: reliability, reasoning coherence, traceability, and controlled evolution over time.

Problem: supervise agents that keep changing

The project frames trust as a longitudinal problem: a medical agent must remain reliable, coherent, and traceable as it learns.

RESEARCH QUESTION

How do we verify that an autonomous medical agent still behaves correctly after tool use, memory updates, and self-improvement cycles?

Autonomy

agents plan task sequences, call tools, and act without direct human intervention

Learning over time

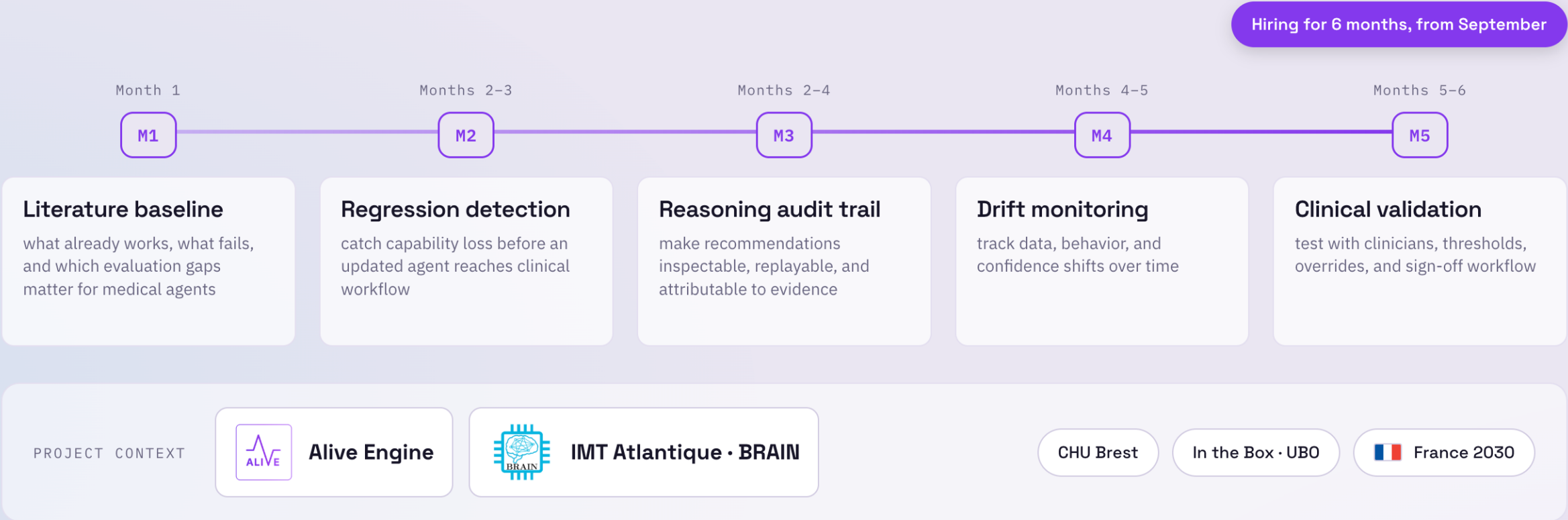
persistent agents change through memory, profiling, and introspection cycles

Medical risk

risk of drift over time, which can affect patient decisions

Project structure

A six-month maturation project with IMT Atlantique BRAIN and Alive Engine, sequencing five supervision modules for learning medical agents.



Current tools miss the learning cycle

The state of the art does not yet supervise whether an autonomous medical agent remains stable after introspection and self-improvement.

01

Chain-of-thought is not enough

It can reduce hallucinations in aggregate, while increasing major omissions in some clinical summarisation settings.

02

Self-improvement can drift

Small updates accumulate. An agent can move away from its original objective without a regression protocol.

03

Observability misses introspection

Guardrails and traces mostly monitor outputs. They rarely verify the agent's self-evaluation cycles.

Medical benchmarks do not test agent stability

Strong medical evaluation datasets exist, but none evaluate whether a learning agent stays reliable over time.

AVAILABLE MEDICAL BENCHMARKS

MedQA

PubMedQA

MedMCQA

MIMIC-IV

BioASQ

MedS-Bench

MISSING FOR AGENTIC AI

- longitudinal stability after memory updates
- traceability across reasoning steps
- behavior drift across interactions
- trust between collaborating agents

Objectives 1–2: prove stability and provenance

The first half of the project turns agent learning into something testable and replayable.

01

Regression detection

Run periodic regression suites after introspection cycles to verify that mastered medical tasks did not degrade.

MedQA · PubMedQA

02

Reasoning audit trail

Attach every medical conclusion to sources, logical steps, and decision points, using knowledge graphs for provenance.

sources · steps · decisions

Objectives 3–4: monitor drift and validate on benchmarks

The second half of the project checks whether supervision works on a concrete medical-agent use case.

03

Behavior drift monitoring

Detect unwanted changes in bias, tone, inter-agent trust, and decision orientation over interactions and learning cycles.

personality · bias · trust

04

Experimental validation

Deploy a medical agent on Alive Engine and evaluate differential diagnosis, drug interaction detection, and literature synthesis.

MIMIC-IV · MedQA · PubMedQA

Thank you

Questions welcome.



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Alive Engine



IMT Atlantique · BRAIN



Alive Engine

The platform used in the project to create, deploy, and supervise persistent agents that learn continuously.

The project uses Alive Engine as the experimental substrate: continuous tasks, persistent state, introspection cycles, sandboxed execution, and real-time supervision.

-  **Persistent memory**
agents consolidate what they learn instead of resetting every run
-  **Local orchestration**
tools, sandboxes, and model calls remain inside the deployment boundary
-  **Audit by default**
every action, evidence source, and handoff can be replayed
-  **Human supervision**
thresholds, approvals, overrides, and halt states are first-class controls



Stroke pathway workspace

● SUPERVISED

objective · reduce dead time from arrival to thrombolysis

5 agents

3 tool scopes

100% actions logged

Clinician workspace

secure thread, console, review queue

Alive Gateway

local endpoint, policy, audit log

Agent runtime

central agent, sub-agents, memory cycles

Hospital tools and data

records, imaging, guidelines, lab systems

Supervision layer

drift, regression, trust scores, sign-off